DEMO 1 · SAP CAP

Build your first SAP CAP app

From an empty folder to a Fiori app. No JavaScript.

DEMO 1 · THE USE CASE

Our customer has a requirement



"We sell to business customers. We need one app to manage our **customers** and their **orders**."

The sales team of a small company

What they need

1. Store customers and Orders
2. Open a customer and see all their orders
3. Know who created or changed a record, and when
4. Every order belongs to one customer

DEMO 1 · THE ARCHITECTURE

Three layers, three folders

FRONTEND

How it looks

Fiori Elements

Ready-made screens, drawn from annotations

`app/`

←

SERVICE

What we show

OData API

CustomerService at /odata/v4/customer

`srv/`

←

DATA MODEL

What we store

CDS entities

Customers and Orders, plus sample data in CSV files

`db/`

←

DATABASE

Where data lives

SQLite

In memory while we develop. CAP creates the tables.

Later: SAP HANA Cloud, same model

Tools: SAP Business Application Studio to write code · cds watch to run it · browser and test.http to test it

DEMO 1 · THE DATA MODEL

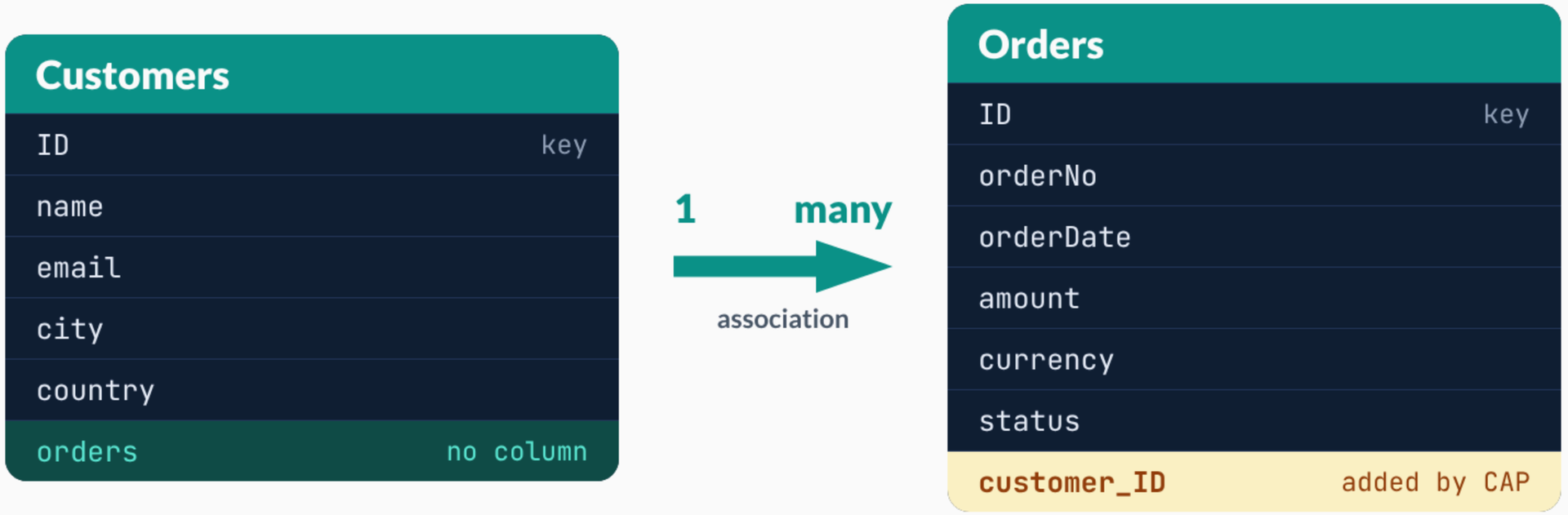
Two entities, two tables

Customers	
ID	key
name	required
email	
city	
country	

Orders	
ID	key
orderNo	required
orderDate	Date
amount	Decimal
currency	default EUR
status	default New

But which customer placed ORD-2026-002?

An association links them



customer_ID stores the ID of the customer who placed the order.

DEMO 1 · THE DATA MODEL

Two aspects, five fields for free

```
using { cuid, managed }
  from '@sap/cds/common';

entity Customers : cuid, managed {
  name : String(100) @mandatory;
  …
}
```

cuid

`ID`

A unique key. CAP fills it with a UUID that never repeats.

managed

`createdAt · createdBy`
`modifiedAt · modifiedBy`

Who created or changed a record, and when.

DEMO 1 · THE SERVICE

A few lines give a full API

srv/customer-service.cds

```
using sales from '../db/schema';

service CustomerService {
  entity Customers as projection on sales.Customers;
  entity Orders    as projection on sales.Orders;
}
```



`/odata/v4/customer`

The service name, without “Service”, in lower case.

Read · Create · Change · Delete

Filter · Sort · Expand

DEMO 1 · SAMPLE DATA

Where every column comes from

sales-Customers.csv

ID | name | email | city | country | createdAt | createdBy | modifiedAt | modifiedBy

sales-Orders.csv

ID | orderNo | orderDate | amount | currency | status | customer_ID | createdAt | createdBy | modifiedAt | modifiedBy

ID from cuid | name our own fields | customer_ID from the association | createdAt from managed

DEMO 1 · TESTING

The browser reads. test.http also writes.

Browser: GET only, so it can only read

test.http: GET and POST, so it can create

Request	Method	Answer
1. Read all customers	GET	200 OK
2. Create an order for Global Logistics	POST	201 Created
3. Create a customer with no name	POST	400 Bad Request

Request 3 is refused because **name** is @mandatory.

Annotations draw the screens

List Report

Name	City	Country
Acme Corp	Berlin	Germany
Global Logistics	Rotterdam	Netherlands
Starlight Retail	Hyderabad	India

`UI.LineItem → the columns`



Object Page

Acme Corp

General Information
orders@acme.example · Berlin · Germany

Orders
ORD-2026-001 · ORD-2026-003

`UI.HeaderInfo → the title`

`UI.FieldGroup + UI.Facets → details`

DEMO 1 · RECAP

About 40 lines of CDS, no JavaScript

- **Data model** in db/: Customers and Orders, linked by an association
- **Database:** SQLite now, SAP HANA Cloud later, same model
- **Service** in srv/: a full OData API at /odata/v4/customer
- **Frontend** in app/: a Fiori app drawn from annotations

Your challenge: give the order page a title, so it shows the order number. Tell me in the comments how it went.